

EN.553.632 Bayesian Statistics

Jonathan Y. Ma

June 2026

Contents

1	Introduction and Probability Review	3
1.1	Bayesian vs Frequentist Perspectives	3
1.2	Bayes' Theorem	3
1.3	Bernoulli and Binomial Models	4
1.3.1	Bernoulli Model	4
1.3.2	Binomial Model	4
1.3.3	Likelihood	4
1.3.4	Maximum Likelihood Estimation	4
1.4	Conjugate Priors: Beta-Binomial Model	4
1.4.1	Posterior Mean	5
1.5	Credible Intervals and HPD Regions	5
1.6	Posterior Predictive Distribution	5
1.7	Exponential Family	6
1.8	Prior Specification	6
1.8.1	Subjective Priors	6
1.8.2	Non-informative Priors	6
1.8.3	Jeffreys Prior	6
1.9	Poisson-Gamma Model	6
1.9.1	Model	6
1.9.2	Prior	6
1.9.3	Posterior	6
1.10	Preview: Computational Methods	7
1.11	Common Distributions and Identities	7
1.11.1	Beta and Gamma Functions	7
1.11.2	Gamma Distribution	7
1.12	Sufficiency and Likelihood Structure	8
1.13	Posterior as a Weighted Average	8
1.14	Bias, Variance, and MSE	8
1.15	Poisson-Gamma Model (Detailed)	9
1.15.1	Likelihood	9
1.15.2	Posterior Derivation	9
1.15.3	Posterior Mean Interpretation	9
1.16	Posterior Predictive Distribution	9
1.17	Exponential Family and Conjugacy	9
1.18	Interpretation of Bayesian Updating	10

2	The Binomial Model	11
2.1	Model Setup	11
2.2	Likelihood and Maximum Likelihood Estimation	11
2.3	Beta Prior	12
2.4	Posterior Distribution	12
2.5	Posterior Summaries	12
2.5.1	Mean	12
2.5.2	Mode	12
2.5.3	Variance	12
2.6	Posterior Predictive Distribution	13
2.7	Credible Intervals	13
2.8	Highest Posterior Density Regions	13
2.9	Shrinkage Interpretation	13
2.10	Rare Event Example	13
2.11	Frequentist vs Bayesian Interpretation	14
3	Poisson Model and Exponential Families	15
3.1	Poisson Model	15
3.2	Likelihood and Sufficiency	15
3.3	Maximum Likelihood Estimation	16
3.4	Gamma Prior	16
3.5	Posterior Distribution	16
3.6	Posterior Summaries	17
3.6.1	Mean	17
3.6.2	Variance	17
3.7	Posterior Predictive Distribution	17
3.8	Example: Birth Rate Comparison	17
3.9	Exponential Family Representation	17
3.9.1	Poisson as an Exponential Family	18
3.10	General Exponential Family Structure	18
3.11	Interpretation and Model Limitations	18

Chapter 1

Introduction and Probability Review

1.1 Bayesian vs Frequentist Perspectives

Definition 1.1.1. Probability can be interpreted in two fundamentally different ways:

- **Frequentist:** Probability is the long-run relative frequency of an event.
- **Bayesian:** Probability is a degree of belief about an uncertain quantity.

In the Bayesian framework, unknown parameters are treated as random variables. This contrasts with the frequentist view, where parameters are fixed but unknown.

Example 1.1.1. Consider a coin flipped $n = 10$ times with 7 heads observed.

- Frequentist estimate: $\hat{\theta} = 0.7$
- Bayesian approach: place a prior $\theta \sim \text{Beta}(\alpha, \beta)$

After observing $y = 7$ heads, the posterior becomes:

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

1.2 Bayes' Theorem

Theorem 1.2.1 (Bayes' Rule). *For parameter θ and data y ,*

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

where

$$p(y) = \int p(y \mid \theta)p(\theta) \, d\theta$$

Remark 1.2.1. The posterior is often written up to proportionality:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

1.3 Bernoulli and Binomial Models

1.3.1 Bernoulli Model

Definition 1.3.1. A Bernoulli random variable satisfies:

$$Y \sim \text{Bernoulli}(\theta), \quad P(Y = 1) = \theta$$

1.3.2 Binomial Model

Definition 1.3.2. For n independent Bernoulli trials,

$$Y \sim \text{Binomial}(n, \theta), \quad P(Y = y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

1.3.3 Likelihood

Given observations $y_1, \dots, y_n$, the likelihood is:

$$L(\theta) = \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}$$

1.3.4 Maximum Likelihood Estimation

Proposition 1.3.1. The MLE for θ in the Bernoulli/Binomial model is:

$$\hat{\theta}_{MLE} = \bar{y}$$

Proof. The log-likelihood is:

$$\ell(\theta) = \sum y_i \log \theta + (n - \sum y_i) \log(1 - \theta)$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - \frac{n - \sum y_i}{1 - \theta}$$

Set equal to zero:

$$\frac{\sum y_i}{\theta} = \frac{n - \sum y_i}{1 - \theta} \Rightarrow \theta = \frac{1}{n} \sum y_i$$

□

1.4 Conjugate Priors: Beta-Binomial Model

Definition 1.4.1. A prior is conjugate if the posterior belongs to the same family as the prior.

Proposition 1.4.1. *If*

$$\theta \sim \text{Beta}(\alpha, \beta), \quad Y \mid \theta \sim \text{Binomial}(n, \theta)$$

then

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

Proof. Posterior:

$$p(\theta \mid y) \propto \theta^y (1 - \theta)^{n-y} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Combine exponents:

$$p(\theta \mid y) \propto \theta^{\alpha+y-1} (1 - \theta)^{\beta+n-y-1}$$

which is the kernel of a Beta distribution. □

1.4.1 Posterior Mean

$$\mathbb{E}[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$$

Remark 1.4.1. This can be interpreted as a weighted average:

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{\alpha}{\alpha + \beta} + (1 - w) \cdot \frac{y}{n}$$

1.5 Credible Intervals and HPD Regions

Definition 1.5.1. An interval (a, b) is a $(1 - \alpha)$ credible interval if

$$\mathbb{P}(a < \theta < b \mid y) = 1 - \alpha$$

Definition 1.5.2. A highest posterior density (HPD) region A satisfies:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$ for $\theta_1 \in A$, $\theta_2 \notin A$

1.6 Posterior Predictive Distribution

Proposition 1.6.1. *For a new observation Z ,*

$$p(Z = 1 \mid y) = \mathbb{E}[\theta \mid y]$$

Proof.

$$p(Z = 1 \mid y) = \int p(Z = 1 \mid \theta) p(\theta \mid y) d\theta = \int \theta p(\theta \mid y) d\theta = \mathbb{E}[\theta \mid y]$$

□

1.7 Exponential Family

Definition 1.7.1. A distribution belongs to the exponential family if it can be written as:

$$p(y | \theta) = f(y)g(\theta) \exp\{\phi(\theta)u(y)\}$$

Remark 1.7.1. Many common distributions (Normal, Binomial, Poisson) belong to this family, which enables conjugate prior constructions.

1.8 Prior Specification

1.8.1 Subjective Priors

Constructed from expert knowledge.

1.8.2 Non-informative Priors

Designed to have minimal influence on the posterior.

Example 1.8.1. A flat prior:

$$\pi(\theta) \propto 1$$

1.8.3 Jeffreys Prior

Definition 1.8.1.

$$\pi(\theta) \propto \sqrt{I(\theta)}, \quad I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log p(y | \theta) \right]$$

Remark 1.8.1. Jeffreys priors are invariant under reparameterization.

1.9 Poisson-Gamma Model

1.9.1 Model

$$Y_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$$

Likelihood:

$$L(\theta) \propto \theta^{\sum y_i} e^{-n\theta}$$

1.9.2 Prior

$$\theta \sim \text{Gamma}(a, b)$$

1.9.3 Posterior

Proposition 1.9.1.

$$\theta | y \sim \text{Gamma}(a + \sum y_i, b + n)$$

1.10 Preview: Computational Methods

In many models, posterior distributions cannot be computed analytically. This motivates:

- Monte Carlo integration
- Importance sampling
- Markov Chain Monte Carlo (MCMC)

These methods will be developed in later chapters.

1.11 Common Distributions and Identities

1.11.1 Beta and Gamma Functions

Definition 1.11.1. The Gamma function is defined as:

$$\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$$

and satisfies:

$$\Gamma(n) = (n-1)! \quad \text{for integer } n$$

Definition 1.11.2. The Beta distribution has density:

$$f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}$$

Proposition 1.11.1. For $\theta \sim \text{Beta}(a, b)$:

$$\mathbb{E}[\theta] = \frac{a}{a+b}, \quad \text{Var}(\theta) = \frac{ab}{(a+b)^2(a+b+1)}$$

1.11.2 Gamma Distribution

Definition 1.11.3. A Gamma distribution with shape a and rate b :

$$f(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}$$

Proposition 1.11.2.

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

Remark 1.11.1. The Gamma distribution is conjugate to the Poisson likelihood.

1.12 Sufficiency and Likelihood Structure

Definition 1.12.1. A statistic $T(y)$ is sufficient for θ if the likelihood can be written as:

$$p(y \mid \theta) = g(T(y), \theta)h(y)$$

Proposition 1.12.1. For:

- *Binomial model: sufficient statistic is $y = \sum y_i$*
- *Poisson model: sufficient statistic is $\sum y_i$*

Remark 1.12.1. This reduction is critical in Bayesian updating since posterior distributions depend only on sufficient statistics.

1.13 Posterior as a Weighted Average

A key Bayesian insight is that posterior estimates combine prior and data:

Proposition 1.13.1. For the Beta-Binomial model:

$$\mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{y}{n}$$

Remark 1.13.1. This shows:

- Prior contributes as pseudo-observations
- Data influence increases with n

1.14 Bias, Variance, and MSE

Definition 1.14.1. Bias of an estimator $\hat{\theta}$:

$$\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$$

Definition 1.14.2. Mean Squared Error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

Remark 1.14.1. Bayesian estimators are often biased but achieve lower MSE through shrinkage.

1.15 Poisson-Gamma Model (Detailed)

1.15.1 Likelihood

$$L(\theta) = \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!} \propto \theta^{\sum y_i} e^{-n\theta}$$

1.15.2 Posterior Derivation

Proof. With prior $\theta \sim \text{Gamma}(a, b)$:

$$\begin{aligned} p(\theta \mid y) &\propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta} \\ &= \theta^{a+\sum y_i-1} e^{-(b+n)\theta} \end{aligned}$$

which is $\text{Gamma}(a + \sum y_i, b + n)$. □

1.15.3 Posterior Mean Interpretation

$$\mathbb{E}[\theta \mid y] = \frac{b}{b+n} \cdot \frac{a}{b} + \frac{n}{b+n} \cdot \frac{\sum y_i}{n}$$

Remark 1.15.1. Again, this is a weighted average of prior mean and sample mean.

1.16 Posterior Predictive Distribution

Definition 1.16.1. The posterior predictive distribution is:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta$$

Remark 1.16.1. This integrates out parameter uncertainty, a key advantage over plug-in estimators.

1.17 Exponential Family and Conjugacy

Definition 1.17.1. A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y) c(\phi) \exp\{\phi t(y)\}$$

Proposition 1.17.1. For exponential families:

- Sufficient statistics appear naturally as $t(y)$
- Conjugate priors exist due to exponential structure

Remark 1.17.1. Examples:

- Binomial $\leftrightarrow$ Beta
- Poisson $\leftrightarrow$ Gamma
- Normal $\leftrightarrow$ Normal / Inverse-Gamma

1.18 Interpretation of Bayesian Updating

Remark 1.18.1. Bayesian inference provides:

- Full uncertainty quantification (posterior distribution)
- Direct probabilistic statements about parameters
- Natural predictive distributions

Example 1.18.1. In rare-event estimation, even if $y = 0$, the posterior remains well-defined:

$$\theta \mid y \sim \text{Beta}(a, b + n)$$

which avoids degenerate estimates like $\hat{\theta} = 0$.

Chapter 2

The Binomial Model

2.1 Model Setup

Definition 2.1.1. Let $Y \mid \theta \sim \text{Binomial}(n, \theta)$ with probability mass function:

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}, \quad \theta \in [0, 1]$$

Remark 2.1.1. If $Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, then $Y = \sum_{i=1}^n Y_i \sim \text{Binomial}(n, \theta)$. The sufficient statistic is $y = \sum Y_i$.

2.2 Likelihood and Maximum Likelihood Estimation

Ignoring constants:

$$L(\theta) \propto \theta^y (1 - \theta)^{n-y}$$

Proposition 2.2.1. *The log-likelihood is:*

$$\ell(\theta) = y \log \theta + (n - y) \log(1 - \theta)$$

Proposition 2.2.2. *The MLE for θ is:*

$$\hat{\theta} = \frac{y}{n}$$

Proof. Differentiate:

$$\frac{d\ell}{d\theta} = \frac{y}{\theta} - \frac{n-y}{1-\theta}$$

Set equal to zero:

$$\frac{y}{\theta} = \frac{n-y}{1-\theta} \Rightarrow \theta = \frac{y}{n}$$

□

2.3 Beta Prior

Definition 2.3.1. Assume prior:

$$\theta \sim \text{Beta}(a, b), \quad \pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$$

Remark 2.3.1. The parameters (a, b) can be interpreted as prior pseudo-counts:

$$a - 1 \text{ successes, } b - 1 \text{ failures}$$

2.4 Posterior Distribution

Theorem 2.4.1. If $\theta \sim \text{Beta}(a, b)$ and $Y \mid \theta \sim \text{Binomial}(n, \theta)$, then:

$$\theta \mid y \sim \text{Beta}(a + y, b + n - y)$$

Proof.

$$\begin{aligned} p(\theta \mid y) &\propto \theta^y (1 - \theta)^{n-y} \cdot \theta^{a-1} (1 - \theta)^{b-1} \\ &= \theta^{a+y-1} (1 - \theta)^{b+n-y-1} \end{aligned}$$

□

2.5 Posterior Summaries

2.5.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

Remark 2.5.1. This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{y}{n}$$

2.5.2 Mode

$$\text{Mode}(\theta \mid y) = \frac{a + y - 1}{a + b + n - 2}$$

2.5.3 Variance

$$\text{Var}(\theta \mid y) = \frac{(a + y)(b + n - y)}{(a + b + n)^2(a + b + n + 1)}$$

2.6 Posterior Predictive Distribution

Definition 2.6.1. For a new observation $\tilde{Y}$:

$$p(\tilde{Y} = 1 \mid y) = \int \theta p(\theta \mid y) d\theta$$

Proposition 2.6.1.

$$p(\tilde{Y} = 1 \mid y) = \mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

2.7 Credible Intervals

Definition 2.7.1. An interval (l, u) is a $(1 - \alpha)$ credible interval if:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

2.8 Highest Posterior Density Regions

Definition 2.8.1. A set A is a HPD region if:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$ for $\theta_1 \in A$, $\theta_2 \notin A$

2.9 Shrinkage Interpretation

Proposition 2.9.1. *The posterior mean can be written as:*

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{a}{a + b} + (1 - w) \cdot \frac{y}{n}$$

where:

$$w = \frac{a + b}{a + b + n}$$

Remark 2.9.1. • Small n : prior dominates

- Large n : data dominates

2.10 Rare Event Example

Example 2.10.1. Suppose:

$$n = 20, \quad y = 0, \quad \theta \sim \text{Beta}(2, 20)$$

Then:

$$\theta \mid y \sim \text{Beta}(2, 40)$$

Remark 2.10.1. The MLE gives $\hat{\theta} = 0$, while the Bayesian estimator remains strictly positive.

2.11 Frequentist vs Bayesian Interpretation

Remark 2.11.1. Frequentist interval:

$$\mathbb{P}(l(Y) < \theta < u(Y) \mid \theta) = 1 - \alpha$$

Bayesian interval:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

Chapter 3

Poisson Model and Exponential Families

3.1 Poisson Model

Definition 3.1.1. A random variable Y follows a Poisson distribution with rate $\theta > 0$ if:

$$Y \mid \theta \sim \text{Poisson}(\theta), \quad p(y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!}, \quad y \in \mathbb{N}$$

Proposition 3.1.1. For $Y \sim \text{Poisson}(\theta)$:

$$\mathbb{E}[Y] = \theta, \quad \text{Var}(Y) = \theta$$

Remark 3.1.1. The equality of mean and variance is a defining feature of the Poisson model. Large deviations between sample mean and variance indicate model misspecification.

3.2 Likelihood and Sufficiency

Let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$. The joint likelihood is:

$$p(y_1, \dots, y_n \mid \theta) = \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!} = c(y) \theta^{\sum y_i} e^{-n\theta}$$

Proposition 3.2.1. The sufficient statistic is:

$$T(y) = \sum_{i=1}^n y_i$$

Remark 3.2.1. All information about θ is contained in the total count $\sum y_i$.

3.3 Maximum Likelihood Estimation

Proposition 3.3.1. *The MLE for θ is:*

$$\hat{\theta} = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Proof. Log-likelihood:

$$\ell(\theta) = \sum y_i \log \theta - n\theta$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - n$$

Set to zero:

$$\theta = \frac{1}{n} \sum y_i$$

□

3.4 Gamma Prior

Definition 3.4.1. Assume prior:

$$\theta \sim \text{Gamma}(a, b), \quad \pi(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}$$

Proposition 3.4.1.

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

3.5 Posterior Distribution

Theorem 3.5.1. *If $\theta \sim \text{Gamma}(a, b)$ and $Y_i \stackrel{iid}{\sim} \text{Poisson}(\theta)$, then:*

$$\theta \mid y \sim \text{Gamma}\left(a + \sum y_i, b + n\right)$$

Proof.

$$\begin{aligned} p(\theta \mid y) &\propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta} \\ &= \theta^{a+\sum y_i-1} e^{-(b+n)\theta} \end{aligned}$$

□

3.6 Posterior Summaries

3.6.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a + \sum y_i}{b + n}$$

Remark 3.6.1. This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{b}{b + n} \cdot \frac{a}{b} + \frac{n}{b + n} \cdot \frac{\sum y_i}{n}$$

3.6.2 Variance

$$\text{Var}(\theta \mid y) = \frac{a + \sum y_i}{(b + n)^2}$$

3.7 Posterior Predictive Distribution

Definition 3.7.1. For a new observation $\tilde{Y}$:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)p(\theta \mid y) \, d\theta$$

Remark 3.7.1. This results in a negative binomial distribution, reflecting overdispersion relative to the Poisson model.

3.8 Example: Birth Rate Comparison

Example 3.8.1. Suppose:

- Group 1: $n_1 = 111$, $\sum y_i = 217$
- Group 2: $n_2 = 44$, $\sum y_i = 66$
- Prior: $\theta \sim \text{Gamma}(2, 1)$

Posteriors:

$$\theta_1 \mid y \sim \text{Gamma}(219, 112), \quad \theta_2 \mid y \sim \text{Gamma}(68, 45)$$

3.9 Exponential Family Representation

Definition 3.9.1. A distribution belongs to the exponential family if:

$$p(y \mid \phi) = h(y)c(\phi) \exp\{\phi t(y)\}$$

3.9.1 Poisson as an Exponential Family

Proposition 3.9.1. *The Poisson distribution can be written as:*

$$p(y \mid \theta) = \frac{1}{y!} \exp\{y \log \theta - \theta\}$$

Remark 3.9.1. This corresponds to:

$$t(y) = y, \quad \phi = \log \theta$$

3.10 General Exponential Family Structure

Definition 3.10.1. A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y)c(\phi) \exp\{\phi t(y)\}$$

Proposition 3.10.1. *For exponential families:*

- $t(y)$ is a sufficient statistic
- Conjugate priors exist due to exponential structure

Remark 3.10.1. Common examples:

- Binomial $\leftrightarrow$ Beta
- Poisson $\leftrightarrow$ Gamma
- Normal $\leftrightarrow$ Normal / Inverse-Gamma

3.11 Interpretation and Model Limitations

Remark 3.11.1. The Poisson model assumes:

- independence of events
- constant rate θ
- equal mean and variance

Remark 3.11.2. If variance exceeds the mean, a negative binomial model may be more appropriate.